Economic complexity – Homework 3

Problem and gap

Skill-relatedness research shows that economies tend to move into activities close to what they already do. Neffke, Otto and Weyh (2017) infer how related industries are from the labor flows between them (revealed relatedness). They find that this skill-relatedness predicts local industry growth better than co-location or value chain relations. Hidalgo et al. (2007) apply the same logic in a network: countries tend to develop products close to those they already make. In music, Percino, Klimek and Thurner (2014) build a similarity network of styles from their instrumentation. They show that a style's instrumentational complexity changes as it attracts more artists, but they do not use artist transitions to measure relatedness. Shi, Lim and Suh (2018) do infer closeness between subgenres from musicians listed in both, but their measure has no timing or direction, and they use it to study how listeners respond to genre-crossing, not to explain which genres grow.

Genre mobility also differs from labor mobility: an artist has no employer to leave, can work in several genres at once, and can enter a new genre with a single release. A genre transition is therefore defined as an artist's first release in a genre they have not released in before, which Discogs release dates would make measurable.

***Gap*:** To our knowledge, no study has tested whether relatedness revealed through artist transitions explains music genre growth, or whether it does so better than instrument-based relatedness. It is therefore unclear whether instruments capture the skills that make a move between genres feasible.

Research question and hypothesis

Research question: Does skill-relatedness measured through artists' transitions between genres predict genre growth better than shared instruments do?

Genre growth is measured as the change in the number of artists releasing in a genre between two periods, relative to all active artists.

Mechanism: Genres form a network in which artists flow along some links and not others. A genre grows when artists can enter it cheaply, which is easier when many artists in large, related genres can step over. Transitions capture everything that makes such a move feasible: playing skills, production know-how, shared labels and scene contacts. Instruments capture only one visible ingredient, and the same instrument can mean very different skills in different genres, like guitar in metal versus in country. A genre's position in the transition network should therefore predict its growth better than its instrument links.

H1: Genres with stronger transition links from large, related genres grow faster in the following period. Transitions are measured in an earlier period than growth, so growth cannot cause the links.

H2: Transition-based relatedness predicts later genre growth better than instrument-based relatedness, especially for genres not defined by their instruments, such as electronic music.

Data and method

Our primary data source would be Discogs. Discogs offers monthly XML dumps of release, artist, label and master release data. These dumps are released under the CC0 license, so the data is free

to use. Each release record provides a year, genre and style tags, artist IDs and credits, and Discogs artist IDs avoid most name-matching problems. Because broad genre categories are too coarse for network construction, the nodes in this analysis represent individual music styles, yielding a few hundred unique nodes.

We would construct two distinct networks on the same nodes. In the transition network (directed, weighted) a directed link from style *A* to style *B* reflects the number of unique artists whose first release in style *B* occurred strictly after their initial release in style *A*. To isolate structural relatedness from simple popularity, these raw transition counts are normalized against a null model based on random expectation given style sizes, following the revealed relatedness methodology. In the instrument network (undirected, weighted) an undirected link between two styles represents the cosine similarity of their instrument profiles, calculated from aggregated instrument credits per style following Percino et al. (2014).

The main measure is relatedness density for each style in both networks: how much of a style's neighborhood, weighted by link strength, is large and active. This is analogous to related employment in Neffke et al. and density in Hidalgo et al. We would regress later growth on each density and compare the two on out-of-sample prediction, so "better" has a concrete meaning. We would also add two descriptive measures, weighted in-degree (how many artists flow into a style) and segregation (whether transition communities match genre families).

For plan B, if release timestamps prove unreliable, the transition network would be replaced by an undirected co-listing network (artists credited across multiple styles within the same timeframe), building on Shi et al. (2018). If instrument credits are too sparse, we would restrict the comparison to styles with enough credit data and say so openly or use a second source such as MusicBrainz for validation.

Expected results and impact

If our hypothesis holds, the empirical analysis would reveal the following patterns:

- transition network: sparse, with strong paths between a few related styles
- predictive comparison: transition-based density predicts later growth better than instrument-based density.
- gap by genre type: it should be largest for styles not defined by their instruments, such as electronic music and hip-hop.
- overlap: the two networks should overlap only partly, meaning instruments capture some of what makes a move feasible but not all of it.
- first descriptive result: how often artists actually release in more than one style.

If instruments predict as well, that is also useful: it would mean a cheap feature-based proxy can replace flow data. The exercise we propose is important for several reasons. For research, it tests whether the skill-relatedness idea from regional economics works in cultural industries, and whether a simple proxy like instruments can replace revealed flows. Flow data is rare in many settings, so a cheap proxy would help a lot if it works. For record labels and A&R teams, it would show which genres artists can credibly enter, so they can plan crossover releases and spot genres likely to grow, instead of relying on instinct. It matters for arts funders and policymakers, as genres with weak links to large neighbors may struggle to attract new artists. Streaming platforms could also use the network to recommend neighboring genres.

References

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